



Customer Churn Analysis –

The dataset contains **7,043 telecom customers**. Overall **churn rate is ~26.5%** (1,869 churned vs. 5,174 retained). Below is a breakdown by different customer attributes:

Demographic Factors

- **Senior Citizens:**
 - Senior → **42% churn**
 - Non-Senior → **24% churn**
 - ◆ **Insight:** Seniors churn almost twice as much as non-seniors.
 - ◆ **Recommendation:** Provide **senior-friendly plans** (lower charges, simplified support).
 - **Dependents:**
 - With Dependents → **15% churn**
 - Without Dependents → **32% churn**
 - ◆ **Insight:** Customers with dependents are more loyal.
 - ◆ **Recommendation:** Offer **family bundles** and highlight savings for dependents.
 - **Partners:**
 - With Partner → **18% churn**
 - Without Partner → **33% churn**
 - ◆ **Insight:** Single customers are more likely to leave.
 - ◆ **Recommendation:** Design **single-user discounts** or loyalty rewards.
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Services & Add-ons

- **Internet Service:**
 - Fiber Optic → **42% churn**

- DSL → **18% churn**
- No Internet → **7% churn**
 - ◆ **Insight:** Fiber optic customers churn the most.
 - ◆ **Recommendation:** Improve fiber service reliability & **offer retention discounts**.
- **Online Security:**
 - Yes → **15% churn**
 - No → **42% churn**
 - ◆ **Insight:** Security services drastically reduce churn.
 - ◆ **Recommendation:** Cross-sell **Online Security bundles** at a discount.
- **Tech Support:**
 - Yes → **16% churn**
 - No → **40% churn**
 - ◆ **Insight:** Customers without tech support leave at 2.5x the rate.
 - ◆ **Recommendation:** Provide **free basic support for first 6 months**.
- **Device Protection:**
 - Yes → **17% churn**
 - No → **38% churn**
 - ◆ **Insight:** Protection services reduce churn by half.
 - ◆ **Recommendation:** Bundle device protection in **premium plans**.
- **Streaming (TV & Movies):**
 - With Streaming → **~30% churn**
 - Without Streaming → **~25% churn**
 - ◆ **Insight:** Entertainment services don't strongly prevent churn.
 - ◆ **Recommendation:** Focus less on streaming, more on **essential services (support, security)**.



Contracts & Billing

- **Contract Type:**

- Month-to-Month → **43% churn**
- One Year → **12% churn**
- Two Year → **3% churn**
 - ◆ **Insight:** Long contracts reduce churn massively.
 - ◆ **Recommendation:** Offer **discounts & free perks** for customers upgrading from monthly to yearly contracts.

- **Payment Method:**

- Electronic Check → **45% churn**
 - Credit Card / Bank Transfer / Mailed Check → **15–20% churn**
 - ◆ **Insight:** E-check users churn more than double compared to others.
 - ◆ **Recommendation:** Encourage **auto-pay via card/bank transfer** with rewards.
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Charges & Tenure

- **Monthly Charges:**

- < \$40 → **15% churn**
- \$40–80 → **25% churn**
- \$80 → **40% churn**
 - ◆ **Insight:** Higher charges = higher churn.
 - ◆ **Recommendation:** Provide **customized discount plans** for high-bill customers.

- **Tenure (Customer Lifetime):**

- < 1 year → **45% churn**
- 1–5 years → **25% churn**

- 5 years → **10% churn**
 - ◆ **Insight:** Early churn is the biggest risk.
 - ◆ **Recommendation:** Focus on **first-year retention programs** (welcome offers, onboarding support).
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Final Recommendations

1. **Focus retention on high-risk groups** → Senior citizens, month-to-month contracts, high monthly charges, and e-check users.
2. **Cross-sell retention-friendly services** → Online security, device protection, and tech support significantly reduce churn.
3. **Strengthen early onboarding** → Customers in their first year are most vulnerable; personalized support here will reduce churn.
4. **Promote long-term contracts** → Use discounts, free months, or bundled perks to move monthly users to yearly/two-year plans.
5. **Improve Fiber Optic service experience** → Since churn is highest here, service reliability & customer support improvements are critical.